Jansen's Leg mechanism

Jansen's Linkage

Jansen's linkage is an 8-bar walking mechanism that was designed by the Dutch artist Theodorus Jansen and used in his kinetic sculptures since 1991. It produces a foot trajectory with a straight line motion in the stride phase (phase of interaction with the ground). The original link lengths available in Theo Jansen's book *The Great Pretender*.

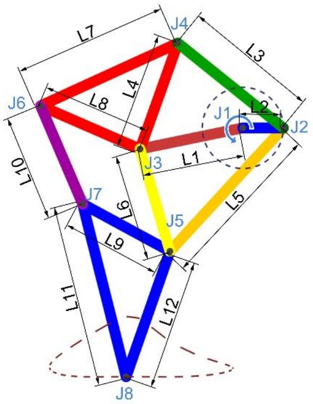

Figure 1: *Jansen's linkage in MechAnalyzer showing its coupler curve*

Figure 2: *Theo Jansen's kinetic sculpture Strandbeest*

Walking Mechanisms

- Biped robots present certain challenges due to the large number of actuators: complexities of design & control, low energy efficiency, overly high price complicating practical use.
- A different way to think about biped robots is to use reduced degrees of freedom by way of using a leg mechanism, which can simulate the walking gait of humans an animals.
- Chebyshev was one of the first to suggest a walking machine based on the fourbar linkage: *the Pantigrade Machine*. (Original models available at the Museum of the History of Physics and Mathematics of Saint Petersburg State University, Peterhof, Russia).
- Chebyshev's machine spawned the development of many different leg mechanisms, using many different standard mechanisms as their bases: *the fourbar linkage with a pantograph, the Watt-II linkage, the Stephenson-III linkage, the Klann linkage, the Jansen linkage etc.*

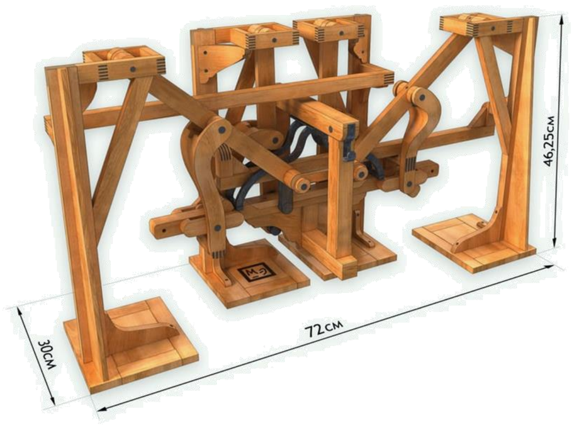

Figure 3: *Chebyshev's Pantigrade Walker*